sMDD (k = ceil(N/2), test: merge_before_connecting) vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_01_11_2026_17-48-09.csv | sMDD: results_python_semi_semi_merge_before_connecting_03_11_2026_14-32-39.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	yes	21	9.49	22.64	1.70×	2.55	5.07	1.88×	872,545	872,545	1.00×	1.64e+06	5.51e+06	2.71×	7,599	7,599	0	0/21/0
	Restricted	yes	156	15.83	25.78	1.51×	4.22	5.88	1.62×	1.77e+06	1.77e+06	0.97×	3e+06	5.39e+06	2.05×	15,583	15,583	0	0/156/0
	RelaxPriority	yes	156	18.27	23.37	1.29×	4.15	5.37	1.42×	1.77e+06	1.75e+06	0.95×	3e+06	4.59e+06	1.80×	591,205	590,478	0	55/101/0
Independent Set	Exact	yes	13	1.34	3.65	1.59×	0.3014	0.3328	0.84×	183,180	163,613	0.66×	213,062	503,285	1.18×	29.08	29.08	0	0/13/0
	Restricted	yes	362	12.06	33.12	2.34×	1.47	1.87	1.10×	1.02e+06	812,049	0.77×	1.07e+06	1.15e+06	0.98×	26.95	22.17	0	0/121/241
	RelaxPriority	yes	359	20.27	31.18	1.42×	2.31	0.9221	0.46×	1.38e+06	527,307	0.43×	1.48e+06	625,252	0.49×	292	309	0	105/61/193
Set Cover	Exact	yes	183	0.5629	9.35	3.51×	0.0598	0.7734	3.72×	39,544	37,825	1.00×	77,884	2.09e+06	6.51×	3.61	3.61	0	0/183/0
	Restricted	yes	2050	9.45	10.13	2.17×	0.3421	0.8480	2.37×	196,896	191,830	1.02×	383,870	1.64e+06	3.25×	8.82	8.82	0	3/2044/3
	RelaxPriority	yes	2027	7.94	22.11	3.14×	0.3041	0.6955	2.26×	161,896	157,227	1.02×	321,861	1.5e+06	3.30×	2.55	2.48	0	16/1851/160
SOC Knapsack	Exact	yes	79	2.14	14.07	3.15×	0.0063	0.9213	23.05×	5,551	4,928	0.96×	10,668	2.69e+06	35.24×	771	771	0	0/79/0
	Restricted	yes	526	0.2641	8.82	7.54×	0.0011	0.4825	32.93×	1,579	4,467	1.89×	2,705	1.36e+06	48.87×	687	772	0	346/179/1
	RelaxPriority	yes	528	1.15	8.68	4.09×	0.0015	0.3476	11.70×	1,860	3,018	1.07×	3,379	1.05e+06	15.58×	891	865	0	247/279/2
Sequencing	Exact	yes	51	5.76	2.23	0.61×	0.6226	0.7131	1.26×	16,373	16,373	1.00×	125,213	125,213	1.00×	1,672	1,672	0	0/51/0
	Restricted	yes	586	9.01	22.85	1.37×	0.3304	1.76	3.08×	15,710	41,192	1.79×	77,051	236,571	2.00×	237,168	230,411	0	291/295/0
	RelaxPriority	yes	446	3.64	32.66	2.80×	0.3194	11.34	7.72×	3,316	10,999	2.14×	63,515	1.8e+06	6.07×	28,780	28,714	0	23/209/214